

JEE MAIN REVISION MASTER: BINOMIAL THEOREM SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 160 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	General Term, Middle Term & Greatest Term	147 PYQs	Expansion Terms
Very High	Properties & Series Sum of Binomial Coefficients	16 PYQs	Coefficient Identities

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
General Term, Middle Term & Greatest Term	$(a + b)^n = \sum_{r=0}^n {}^nC_r a^{n-r} b^r$, $T_{r+1} = {}^nC_r a^{n-r} b^r$, $T_{\frac{n}{2} + 1}$ (if n is even)
Properties & Series Sum of Binomial Coefficients	$\sum_{r=0}^n {}^nC_r = 2^n$, $C_0 - C_1 + C_2 - \dots = 0$, $\sum_{r=1}^n r {}^nC_r = n 2^{n-1}$, $\sum_{r=0}^n ({}^nC_r)^2 = {}^{2n}C_n$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 General Term, Middle Term & Greatest Term (Very High Yield — 147 PYQs)

Core Principle: Expansion of $(x + a)^n$, formula for general $(r+1)^{\text{th}}$ term, finding term independent of x , and middle term determination.

■ **Caution / JEE Pitfall:** In $(a + b)^n$, total number of terms is $n + 1$. If n is odd, there are TWO middle terms: $(\frac{n+1}{2})^{\text{th}}$ and $(\frac{n+3}{2})^{\text{th}}$.

■ **Speed Trick:** In expansion of $(ax^p + bx^q)^n$, the term independent of x occurs at index $r = \frac{np}{p+q}$ (provided r is an integer).

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "If $\frac{2^3}{3}({}^{12}C_2) + \frac{2^5}{5}({}^{12}C_4) + \frac{2^7}{7}({}^{12}C_6) + \dots + \frac{2^{13}}{13}({}^{12}C_{12}) = 3^{13} - \lambda$..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3 + (1+x)^4 + (1+x)^5 + \dots + (1+x)^{99} + (1+kx)^{100}$, $x \dots$ " [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "Let $S = (1/25!) + (1/3!23!) + (1/5!21!) + \dots$ up to 13 terms. If $13S = \frac{2^k n!}{k!}$, $k \in \mathbb{N}$, then $n+k$ is equal to..." [Ans: N/A]

3.2 Properties & Series Sum of Binomial Coefficients (Very High Yield — 16 PYQs)

Core Principle: Combinatorial identities, summing coefficient series via differentiation and integration of $(1+x)^n$.

■ **Caution / JEE Pitfall:** To find the sum of all coefficients of a polynomial expansion $P(x)$, simply substitute $x = 1$.

■ **Speed Trick:** Useful ratio identity: $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$.

Representative Source PYQs:

- **[2026 Evening Shift] Q#2 (MCQ):** "If for $3 \leq r \leq 30$, $({}^{30}C_{(30-r)} + 3({}^{30}C_{(31-r)}) + 3({}^{30}C_{(32-r)}) + ({}^{30}C_{(33-r)}) = {}^m C_r$, then m equals :..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#4 (MCQ):** "The sum of the coefficients of x^{499} and x^{500} in $(1+x)^{1000} + x(1+x)^{999} + x^2(1+x)^{998} + \dots + x^{1000}$ is :..." **[Ans: N/A]**
- **[2025 Evening Shift] Q#2 (MCQ):** "If $\sum_{r=0}^{10} \left(\frac{10!}{r!10^{10-r}} \right) \cdot {}^{11}C_{(r+1)} = \frac{(\alpha)^{11} - 11^{11} 10^{10}}{10!}$, then α is equal to :..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Let the smallest value of $k \in \mathbb{N}$, for which the coefficient of x^3 in $(1+x)^3 + (1+x)^4 + (1+x)^5 + \dots + (1+x)^{99} + (1+kx)^{100}$, $x \neq 0$, is $(43n + (101/4))({}^{100}C_3)$ for some $n \in \mathbb{N}$, be p . Then the value of $p+n$ is :

- (A) 10
- (B) 11 **[Correct]**
- (C) 12
- (D) 13

Correct Answer: B

Solution Analysis: Coefficient of x^3 in the given expression $(1+x)^3 + (1+x)^4 + \dots + (1+x)^{99} + (1+kx)^{100}$ is obtained by adding the coefficients of x^3 from each term. For $(1+x)^r$, coefficient of x^3 is ${}^r C_3$. So, from $(1+x)^3 + (1+x)^4 + \dots + (1+x)^{99}$ the total coefficient of x^3 is $\sum_{r=3}^{99} {}^r C_3 = 3 {}^{100} C_3$. Also, in $(1+kx)^{100}$, coefficient of x^3 is ${}^{100} C_3 k^3$. Hence total coefficient of x^3 is $\sum_{r=3}^{99} {}^r C_3 + {}^{100} C_3 k^3 = 3 {}^{100} C_3 + {}^{100} C_3 k^3$. Now use the standard NCERT identity: $\sum_{r=3}^{99} {}^r C_3 = {}^{100} C_4$. So coefficient becomes ${}^{100} C_3 (3 + k^3)$.